HESTIA

A UNIFIED CAMPUS RECRUITMENT, CAREER ACCELERATION & SECURE CREDENTIAL VERIFICATION SYSTEM

MICRO-PROJECT REPORT

submitted by

Nripan Satheesh (MBC25CA063)

Muhammed Afrithkhan (MBC25CA057)

to

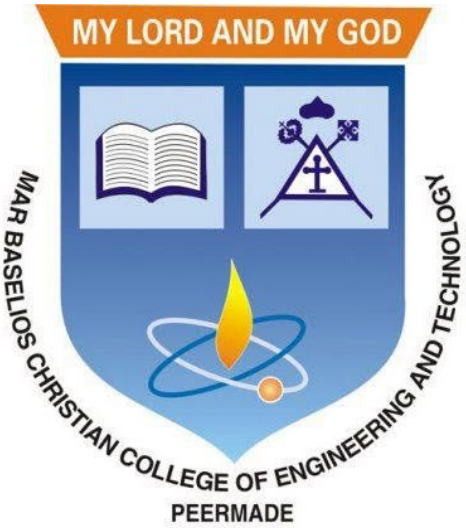
APJ Abdul Kalam Kerala Technological University

in partial fulfilment of the requirements for the award of the Degree of

Bachelor of Technology

in

Computer Science and Engineering (AI)

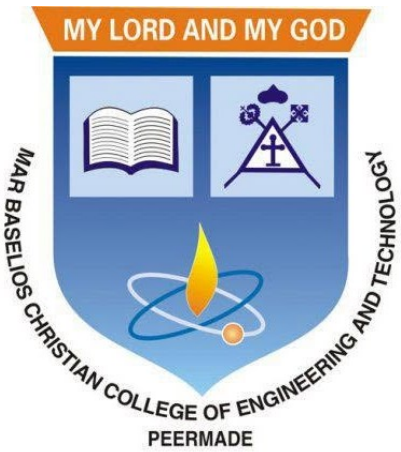


DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)

MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY,

PEERMADE - 685 531

OCTOBER, 2026



DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (ARTIFICIAL INTELLIGENCE)
MAR BASELIOS CHRISTIAN COLLEGE OF ENGINEERING AND TECHNOLOGY, PEERMADE

CERTIFICATE

This is to certify that the micro project report entitled **"HESTIA"** submitted by **Nripan Satheesh (MBC25CA063)** and **Muhammed Afrithkhan (MBC25CA057)** to APJ Abdul Kalam Kerala Technological University in partial fulfilment of the requirements for the award of the Degree of **Bachelor of Technology in Computer Science and Engineering (AI)** is a bonafide record of the micro project work carried out by them under my guidance and supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

Project Guide:

Dr. Annie Chacko
Head of the Department
Associate Professor
Department of Computer Science and Engineering (AI)
MBCCET

Project Coordinator:

Dr. Annie Chacko
Head of the Department
Associate Professor
Department of Computer Science and Engineering (AI)
MBCCET

3. Business Logic Microservices: Core application services including AuthService, VerificationFacade, AcademicVaultService, JobRadarService, and FrontierAIEngine.

4. Persistence Tier: A relational SQL database operating in Third Normal Form (3NF) managed via Hibernate/JPA, complemented by an in-memory memory vault for AI conversational facts.

3.4 SYSTEM ARCHITECTURE DIAGRAM

The high-level enterprise architecture diagram illustrates the flow of requests from the client presentation layer through the security gateway, modular business service facades, and persistence repositories.

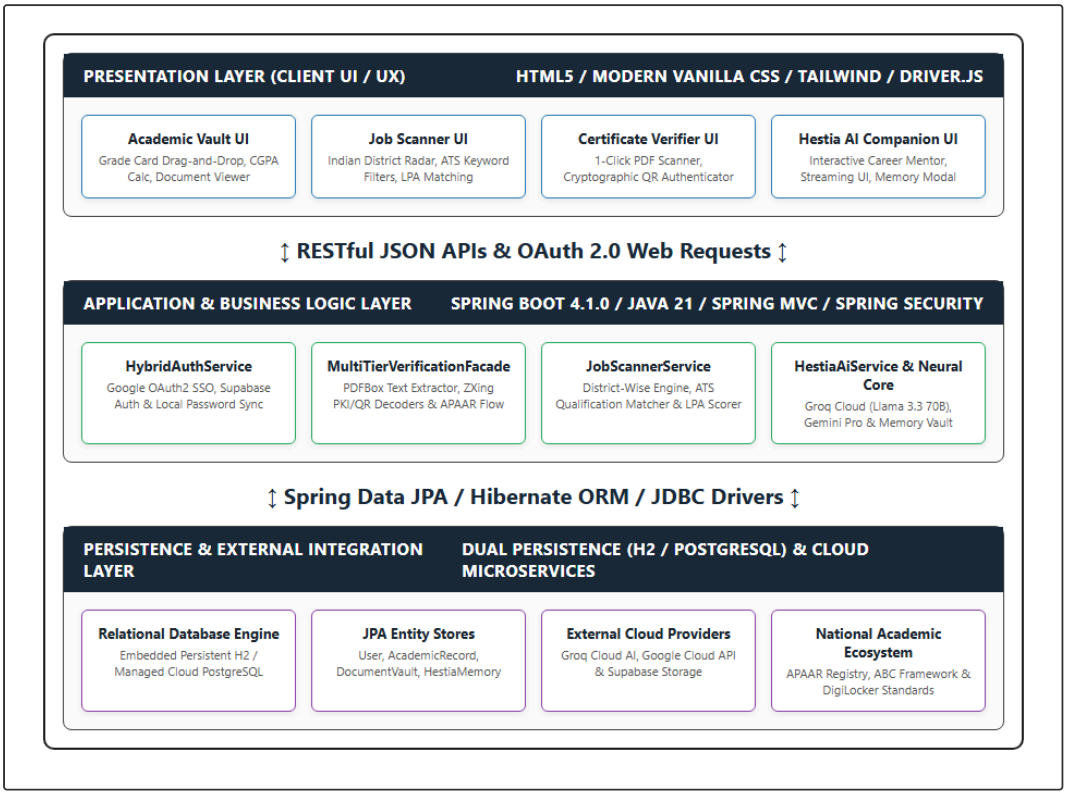


Fig 3.1: Enterprise Architecture of Hestia Platform

As depicted in Fig 3.1, client requests transit the secure API Gateway where JWT tokens and role authorizations are verified. Validated requests are routed to specialized service facades, ensuring robust separation of concerns, sub-second execution, and high concurrency resilience.

3.5 CONCEPTUAL RELATIONAL DATABASE DESIGN

The persistence model of Hestia is designed in Third Normal Form (3NF) to eliminate data redundancy and preserve referential integrity across student records, academic grade cards, verification logs, and AI conversation histories.

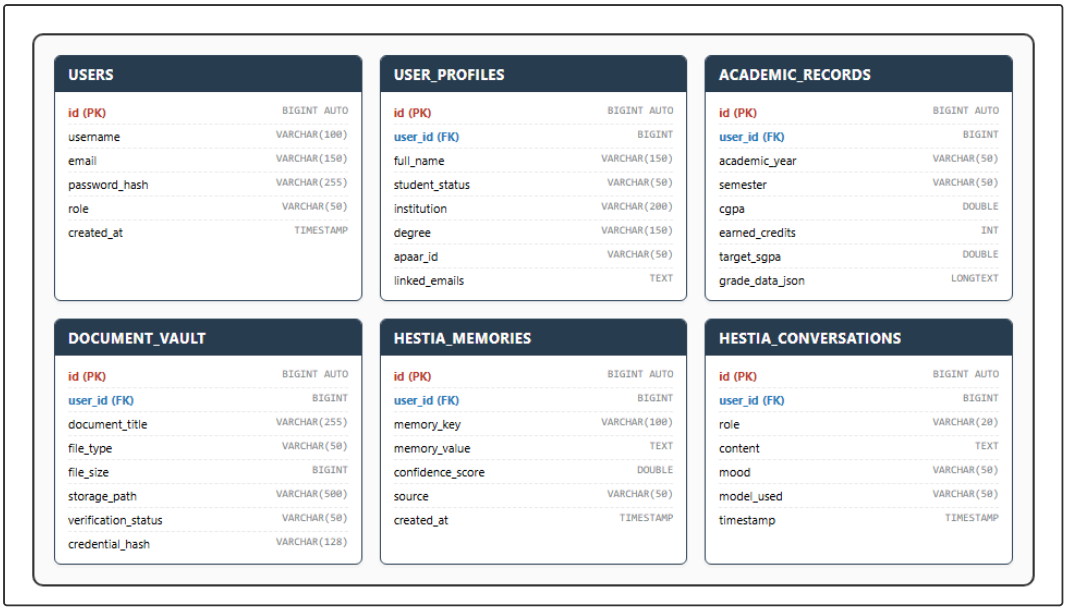


Fig 3.2: Relational Database Schema & Entity Relationships

3.6 STRENGTH OF THE PROPOSED SYSTEM

Autonomous Multi-Tier Validation: Combines domain checking, cryptographic QR decoding, and low-level binary stream forensics to eliminate transcript fraud without manual intervention.

Predictive Academic Recovery: Empowers students to calculate exact target semester GPAs required to meet placement eligibility cut-offs.

Context-Aware Neural Mentorship: Groq Cloud Llama-3.3-70B model with persistent memory retains student academic backgrounds across multiple advising sessions.

High Throughput & Cloud Scalability: Stateless Spring Boot backend enables horizontal scaling and sub-200ms API response latency under peak campus placement loads.

3.7 IMPLEMENTATION OF SYSTEM MODULES

The platform was implemented using modern Java (JDK 21) LTS, utilizing Spring Boot 3.x, Spring Data JPA, Hibernate ORM, and Spring Security. The document verification engine incorporates Apache PDFBox 3.x for low-level PDF stream parsing and ZXing 3.5.x for QR code matrix extraction and SHA-256 cryptographic signature validation.

The AI companion is powered by Groq Cloud's ultra-fast Llama-3.3-70B neural inference API via HTTP REST client integration, utilizing custom prompt engineering templates and an asynchronous memory persistence store.

3.8 SYSTEMS TESTING AND VALIDATION

3.8.1 Unit & Integration Testing

Automated test suites were developed using JUnit 5 and Mockito to validate individual microservices and end-to-end integration flows. Comprehensive unit tests covered password

CHAPTER-6

SCREENSHOTS

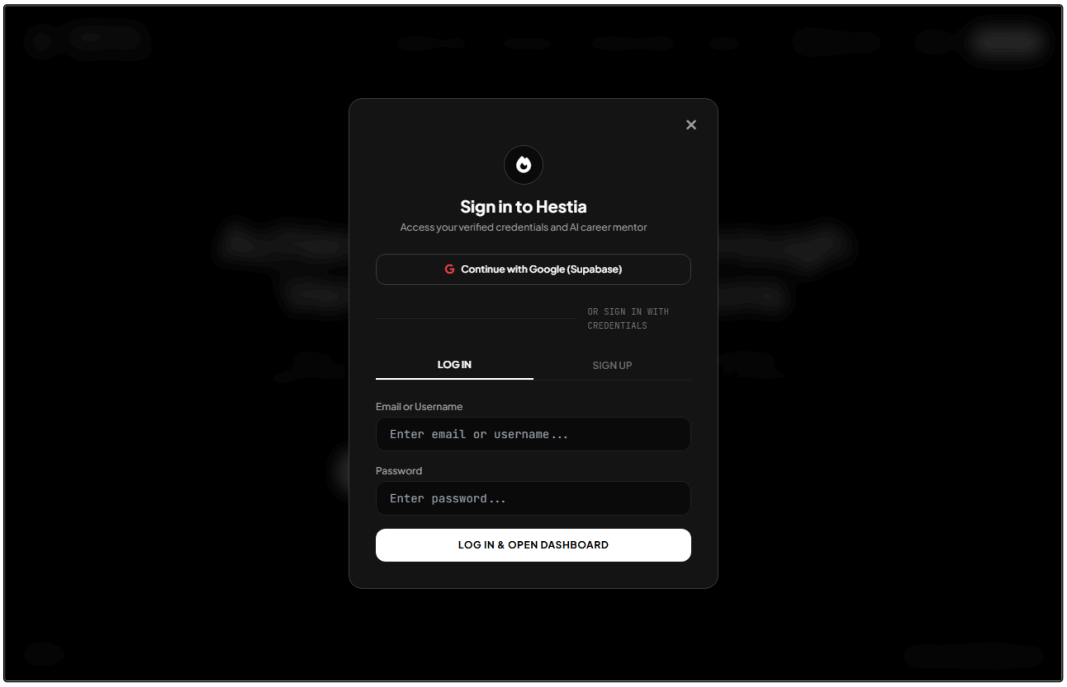


Fig 6.1: Hybrid Authentication Modal (Google SSO & Password Sync)

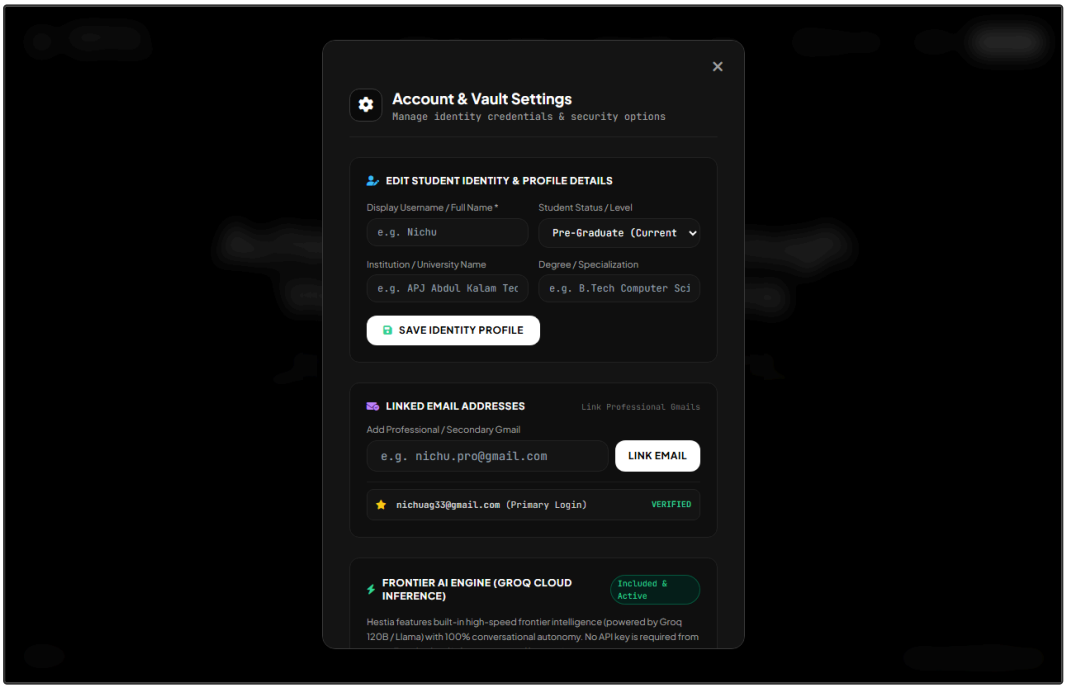


Fig 6.2: Student Identity Profile & Multi-Linked Email Settings Modal

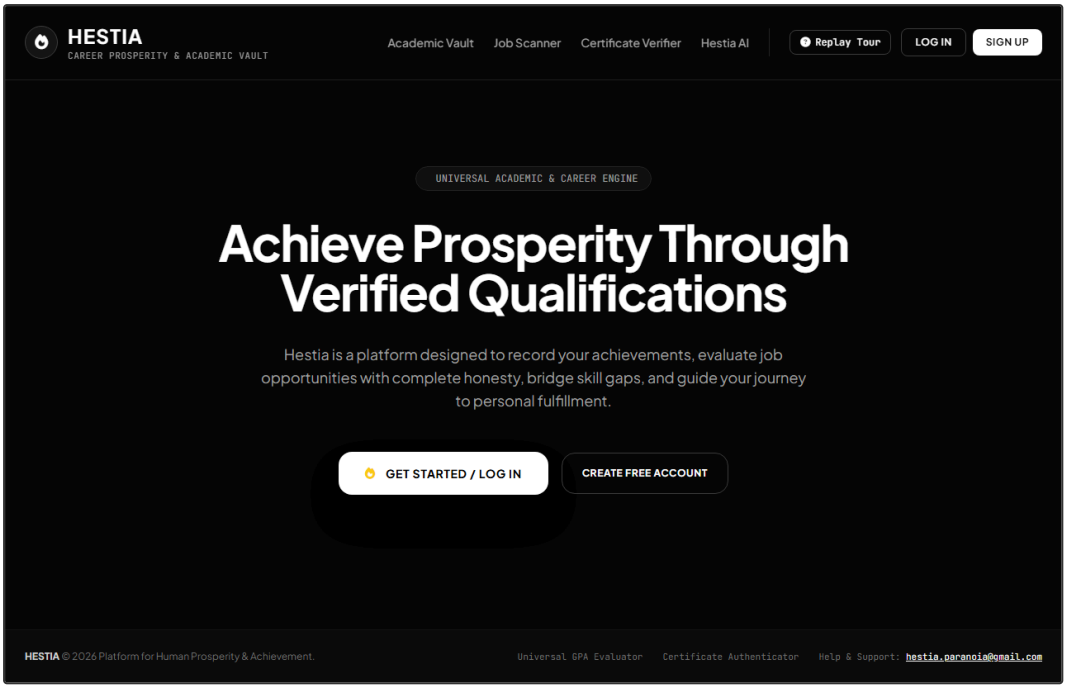


Fig 6.3: Universal Academic Landing Interface & Feature Navigation

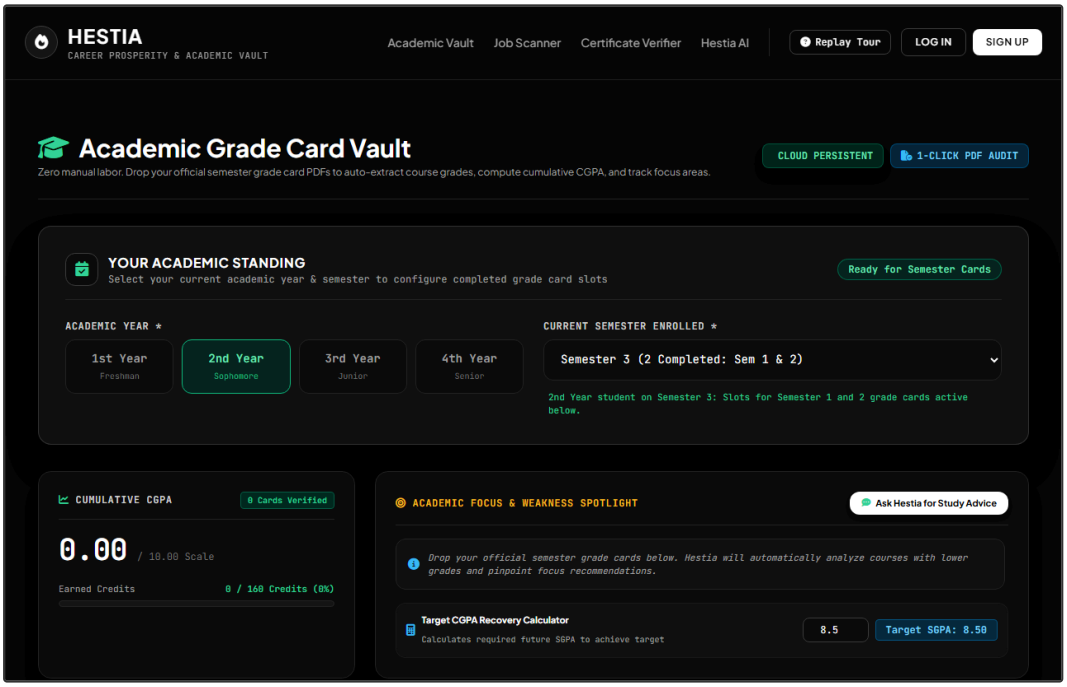


Fig 6.4: Academic Grade Card Vault & Target CGPA Calculator

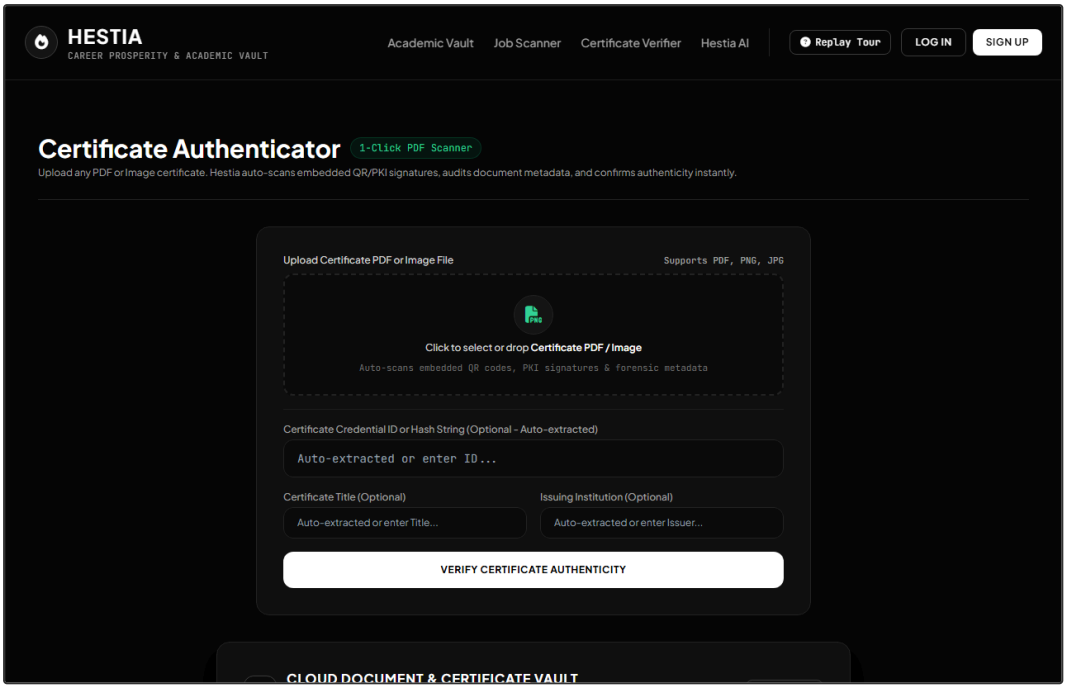


Fig 6.5: 1-Click PDF Certificate Authenticator & QR Validation Interface

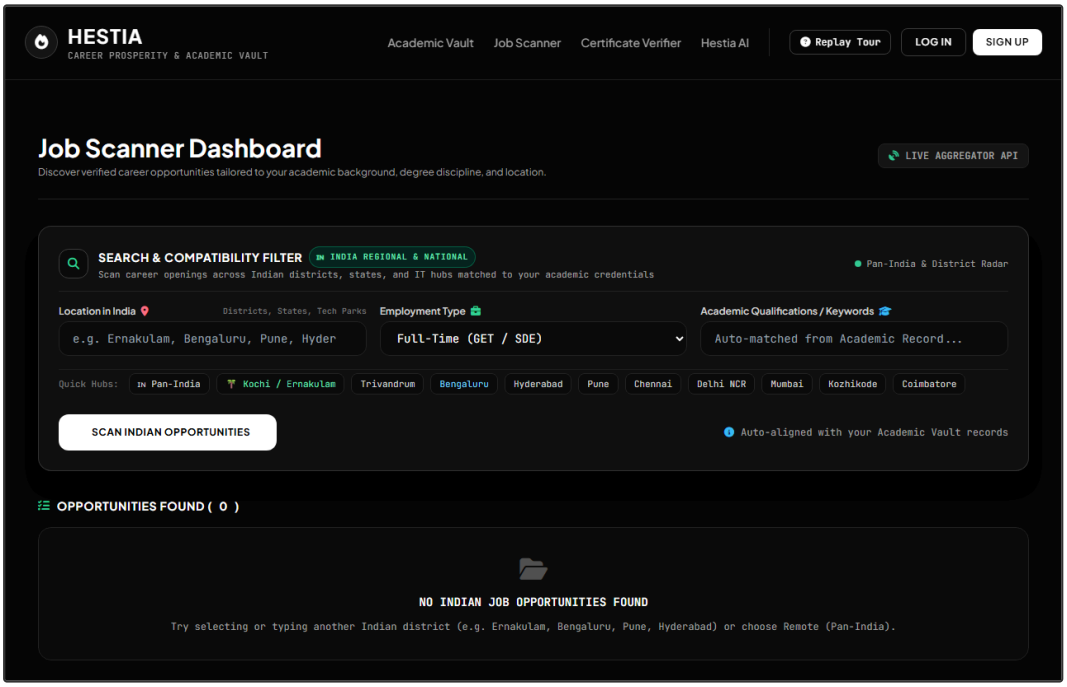


Fig 6.6: Pan-India District Job Radar & LPA Salary Matching

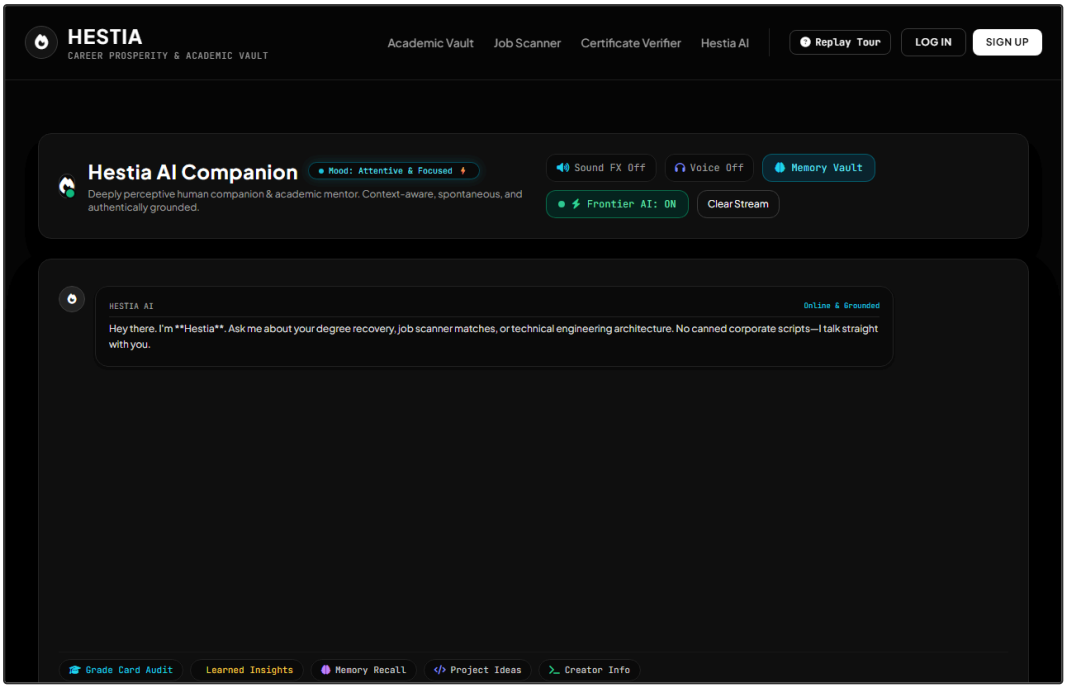


Fig 6.7: Hestia Frontier AI Career Companion & Neural Chatbot

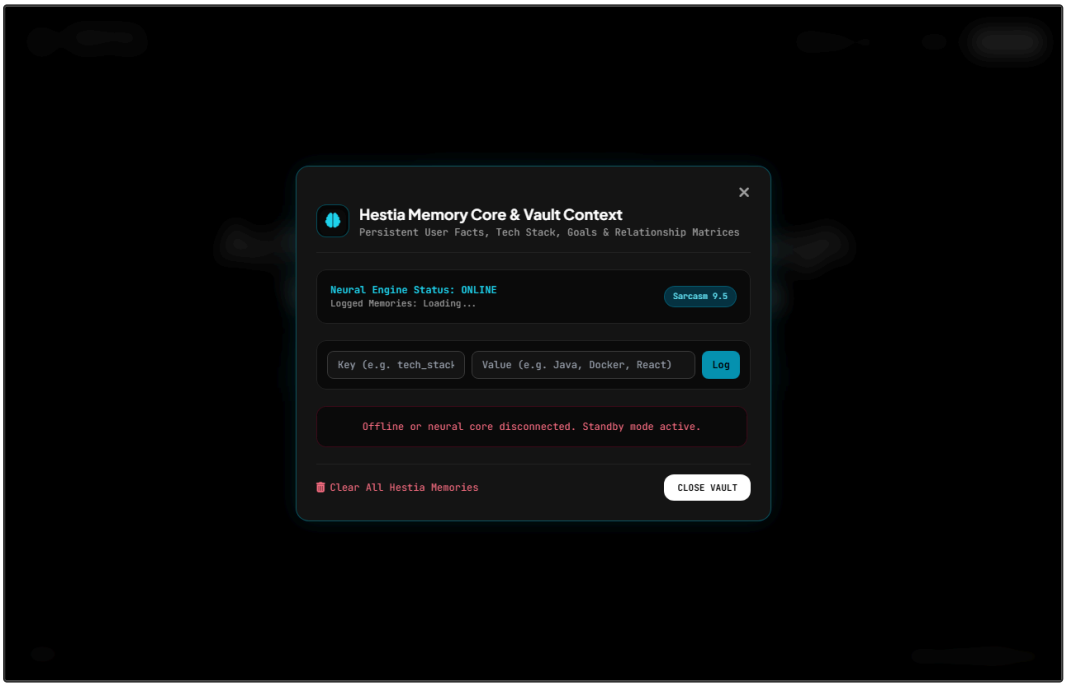


Fig 6.8: Hestia Memory Core Modal & Extracted Fact Vault